

This API Endpoint Plan defines all routes required for RaceDay this ensures that organisers and participants can manage events, categories, enrolments and results in line with the approved functional requirements.

HTTP Method	Route	Description	Role Required	Request Body / Parameters	Expected Response
POST	/api/auth/register	Register a new User account (Organiser or Participant).	Public	{"firstName", "lastName", "email", "password", "phoneNumber", "role"}	201 Created
POST	/api/auth/login	Authenticate user and return access token.	Public	{"email", "password"}	200 OK
GET	/api/profile	Retrieve the logged-in user profile (including associated Participant info if applicable).	Logged-in User	None	200 OK
PUT	/api/profile	Update the logged-in user's profile information.	Logged-in User	{"firstName", "lastName", "phoneNumber"}	200 OK
POST	/api/events	Create a new event.	Event Organiser	{"name", "description", "date", "location", "startTime", "routemap", "eventType", "registrationDeadline"}	201 Created
GET	/api/events	Retrieve all available events.	Logged-in User	None	200 OK
GET	/api/events/{id}	Retrieve details of a specific event.	Logged-in User	None	200 OK
PUT	/api/events/{id}	Update an existing event.	Event Organiser	{"name", "description", "date", "location", "startTime", "routemap", "eventType", "registrationDeadline"}	200 OK
DELETE	/api/events/{id}	Delete an existing event.	Event Organiser	None	204 No Content

HTTP Method	Route	Description	Role Required	Request Body / Parameters	Expected Response
POST	/api/events/{eventId}/categories	Create a category/distance option for an event (MaxParticipants).	Event Organiser	{"categoryName", "distance", "categoryLevel", "entryFee", "maxParticipants"}	201 Created
GET	/api/events/{eventId}/categories	View all categories/distances available for a specific event.	Logged-in User	None	200 OK
POST	/api/enrolments	Register a participant into a specific event category (PARTICIPANT_EVENT).	Participant	{"eventId", "categoryID"}	201 Created
GET	/api/events/{eventId}/enrolments	View all participant enrolments for a specific event.	Event Organiser	None	200 OK
POST	/api/results	Record a participant's finishing time, position, and status (RESULT).	Event Organiser	{"participant_eventID", "time", "position", "status"}	201 Created
GET	/api/results/my	View the logged-in participant's own race results and history.	Participant	None	200 OK
GET	/api/participants/me/performance	Retrieve historical performance analytics (PERFORMANCE_HISTORY).	Participant	None	200 OK

Notes:

- The API architecture mirrors the domain constraints and relationships established in the ERD, ensuring that routes like `/api/events/{eventId}/categories` and `/api/enrolments` directly enforce database integrity between events, participants, and category allocations.
- Authentication is handled at the application layer to issue access tokens, which correctly separates session management logic from the core domain entities represented in the ERD.
- Future enhancements—such as live weather tracking and advanced route integrations—are kept outside the current scope of core functional requirements.